(Please write your Enrollment Number)

Enrollment No. 0160117202 1

**MID-TERM EXAMINATION**
**Course Name: B. Tech: AI and DS, Semester:<6>**
**(March, 2024) OFFLINE mode**

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| **Subject Code: BAI- 302** | **Subject: Natural Language Processing** |
| **Time :1 ½Hours** | **Maximum Marks : 30** |
| **Note:Q. 1 is compulsory.** | |

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| **Q1** | | **(2.5*4)** | |
| | (a) Discuss difference between Lemmatization and Stemming. | UNIT-1 | $CO_1$ |
| | (b) How does context-free grammar facilitate syntactic analysis in NLP systems? | UNIT-1 | |
| | (c) What is Parsing in regards to NLP and make parse tree for these sentences<br>"The dog barks loudly."<br>"She reads a book in the library." | UNIT-2 | $CO_2$ |
| | (d) What is Named Entity Recognition (NER) and what role does it play in natural language processing tasks? | UNIT-2 | |
| **Q2** | **(Attempt any Two Parts ) UNIT-1 ($CO_1$)** | | **(5,5)** |
| | (a) Explain different phases of NLP with example of each phase. | | |
| | (b)Explain tokenization is in the context of natural language processing and different type of tokenization and apply each type of tokenize on given paragraph.<br><br>Education is the cornerstone of progress, nurturing minds and fostering critical thinking. It empowers individuals with skills and knowledge to navigate challenges and contribute to society. Investing in education is paramount for building a brighter and more inclusive future. | | |
| | (c) What is meant by Ambiguity and Write about different type of ambiguity with example? | | |
| **Q3** | **(Attempt any Two Parts ) UNIT-2 ($CO_2$)** | | **(5,5)** |
| | (a) Create Bag of Words (BOW) Vectors and TF-IDF before cleaning and after cleaning text<br>S1: "I explore Data Science methods."<br>S2: "I enjoy Machine Learning algorithms."<br>S3: "I explore and enjoy into Deep Learning." | | |
| | (b) Write in detail about Glove model and Find word-word co-occurrence matrix(x) of thefollowing sentence.<br>S1:"I enjoy watching a movie."<br>S2:"I like going to the cinema."<br>S3: "I prefer watching films." | | |
| | (c) ExplainWord2vec Word embedding and also types in details with suitable example. | | |